E0X Calculating residence time distribution quantities

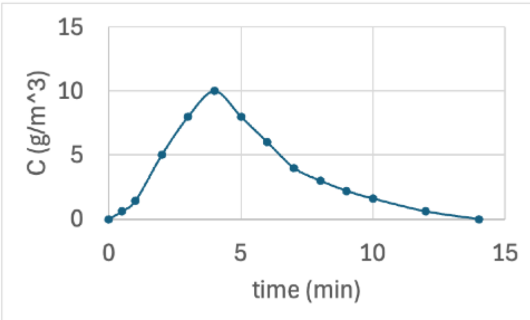
This is Fogler Examples 16-1 and 16-2.

A sample of the tracer hytane was injected as a pulse into a reactor and the exit concentration was measured as a function of time. The data is given below.

1. Construct a curve showing C(t).
2. Construct the E(t) curve.
3. Construct the F(t) curve.
4. Determine the mean residence time.
5. Calculate the fraction of fluid that spends between 3 and 6 minutes in the reactor.

t (min)	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
C (g/m^3)	0	0.6	1.4	5	8	10	8	6	4	3	2.2	1.6	0.6	0

①

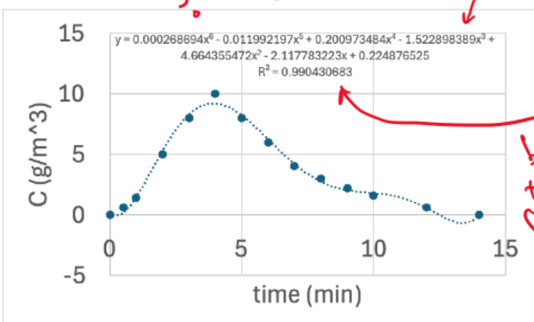


here we just plot the data in the table

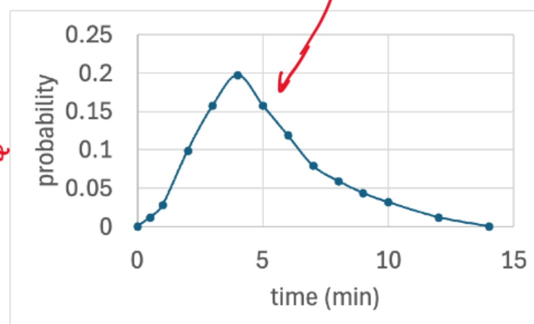
② $E(t) = \dfrac{C(t)}{\int_0^\infty C(t)\,dt}$

integrals aren't sensitive to "overfit" polynomials. Use a lot of sig figs to avoid errors.

$\dfrac{C(t)}{50.6 \frac{g \cdot min}{m^3}}$

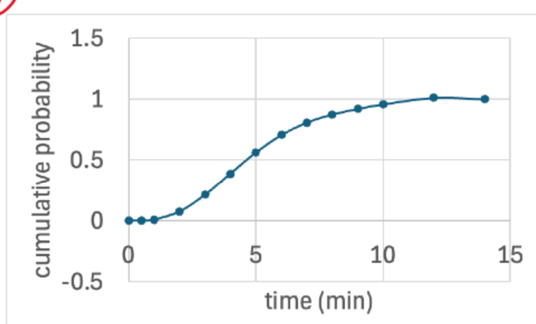


integrate btwn 0 & 14 to approximate $\int_0^\infty C(t)\,dt$



$\int_0^{14} y\,dt \approx \int_0^\infty C(t)\,dt = 50.6 \frac{g \cdot min}{m^3}$

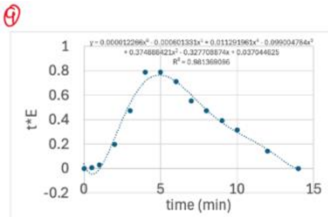
③



$F(t) = \int_0^t E(t)\,dt$.

this plot is $\int_0^t y\,dt$ for each t value in the table

④



$t_m = \int_0^\infty t\,E(t)\,dt$.

new data table

t	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
$t \cdot E(t)$	0	0.006	0.027	0.20	0.47	0.79	0.79	0.71	0.55	0.47	0.39	0.32	0.14	0

plot new data, fit using lots of sig figs in your coefficients, $\int_0^{14} y\,dt$. Get $t_m = 5.2$ min

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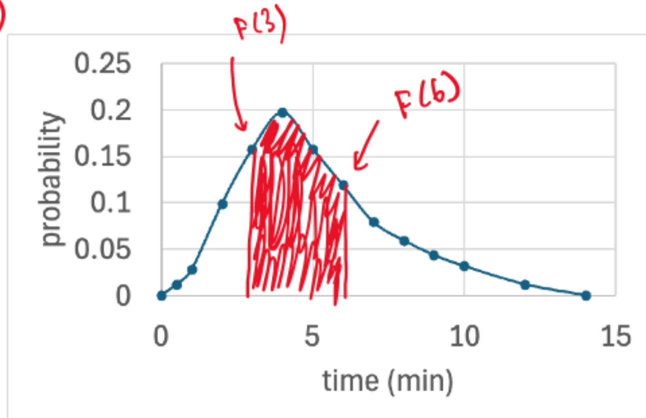

integrate E(t) btwn 3 & 6 minutes to get fraction.

$$\int_3^6 y\,dt = F(6) - F(3) = 0.494$$

so 49% of molecules have τ btwn 3 & 6 minutes